

Results Draft: Embodied Reasoning as Grounded Selection

Anonymous Submission

Anonymous Affiliation

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Results

Embodied Reasoning

Evaluation protocol and evidential order. We evaluate at anticipation points from the Ego4D Goal-Step held-out split. The frozen world model receives the observation and completed-action history, then proposes a shuffled Top- K action boundary; its scores and ranks are hidden from the policy. Unless noted otherwise, all comparisons use the same first-person evaluation cohort and the common covered set ($a_t^{\text{GT}} \in D_t$; $n=1,000$). We organize the evidence from outcome to mechanism: first the controlled selection comparison, then capability decomposition, and finally paired interventions on history and task belief. This order separates an improvement in *what is selected* from a superficial change in *how the trace is written*.

Candidate Alignment, Not Answer-Only Supervision, Improves Selection

The primary comparison controls backbone, examples, training budget, and evaluation prompts; the only change is whether training exposes the world-model candidate set. Candidate alignment (θ_{CE}) reaches 28.8% selection accuracy, compared with 21.9% for answer-only supervision (GT-only). On the official non-malformed paired set ($n=948$, 85 video clusters) their difference is +7.70 pp, with a video-cluster bootstrap 95% CI [+3.73, +11.93]. In contrast, GT-only differs from the unaligned policy by only +0.9 pp (95% CI [-2.0, +3.7]). Candidate alignment also exceeds the frozen world-model Top-1 policy (28.8% vs. 24.2%; +4.6 pp strictly); the official paired gate gives +6.06 pp (95% CI [+0.87, +11.22]; $n=957$, 86 clusters). Thus, the gain cannot be explained by additional exposure to the correct answer or by copying the world-model ranking: it requires learning to discriminate among visually plausible alternatives. Figure 1b traces SelAcc along the branching training path: candidate alignment is the step that carries the policy across the world-model reference.

Where the Gain Appears: Five Capability Axes

Aggregate accuracy alone does not reveal whether the policy preserves a correct predictive prior or repairs an incorrect one. We therefore split selection into G_1 retention (world-model Top-1 correct) and G_2 correction (world-model Top-1 wrong). As shown in Figure 1a and Table 1, candidate alignment raises G_1 from 29.7% to 37.4% and G_2 from 18.7% to 27.8%. GT-only instead lowers G_1 to 26.5%, even though it slightly raises G_2 to 20.8%. The final EGO model further raises G_1 to 42.5%, while G_2 settles at 25.6%. Providing a predictive boundary at inference is therefore insufficient: the policy must be trained to recognize when that boundary’s leading hypothesis should be kept and when it should be overridden.

The same pattern appears on trajectory continuity. Ground-truth actions that continue the immediately preceding behavior constitute 31.0% of the evaluation set. Candidate alignment recovers 43.5% of them, compared with 28.4% for the base and 32.9% for GT-only, while the final EGO model reaches 48.4%. Continuation precision stays within 68–73% across evaluated model states. The comparatively stable precision rules out a looser tendency to repeat the previous action: alignment primarily improves recall of a high-precision temporal cue.

Causal Test: The Learned Advantage Is Carried by History

We next remove completed-action history while holding the image, candidate set, and checkpoint fixed. Candidate alignment loses 10.1 pp (28.8% $\rightarrow$ 18.7%; 95% CI [+6.8, +13.3]), whereas the base and GT-only losses are 2.9 pp and 4.5 pp, respectively, and the replay checkpoint loses 6.4 pp; the intervention was not re-run for the final checkpoint. More importantly, without history, candidate alignment is indistinguishable from both the base (+0.6 pp, 95% CI [-1.8, +3.0]) and GT-only (+1.3 pp, 95% CI [-1.2, +3.9]). These paired estimates quantify the collapse. If training had merely installed a better action prior, its advantage would remain after history was masked. Instead, the advantage vanishes, indicating that candidate alignment teaches the model to read the trajectory and use it to choose

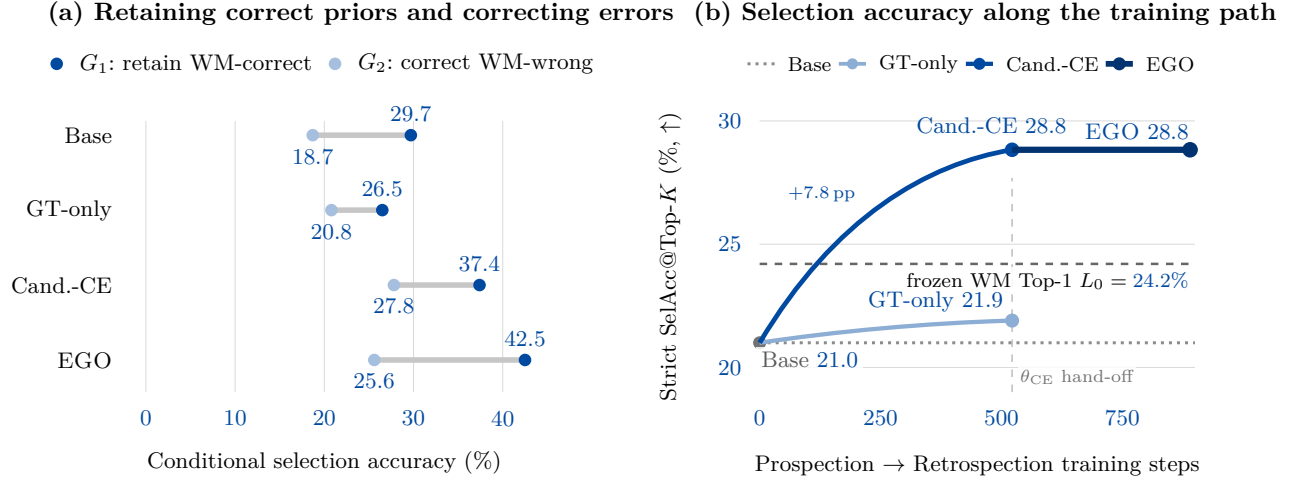


Figure 1: **Candidate alignment teaches grounded selection rather than an answer prior.** (a) G_1 evaluates retention where the world-model Top-1 is correct, whereas G_2 evaluates correction where it is wrong. Candidate alignment improves both; GT-only erodes G_1 despite seeing the correct answers, while the final EGO model reaches 42.5% retention. (b) Strict SelAcc@Top- K against training steps along the branching training path. All arms share the same frozen origin (21.0%). Over the Prospection phase (522 selection-CE steps) candidate alignment reaches 28.8% whereas the answer-only control reaches only 21.9%. EGO then branches from the θ_{CE} adapter and trains a further 368 steps of Retrospection-with-Replay ($\rho=0.15$), holding 28.8%. Base is frozen and is therefore constant by construction. Lines are guides to the eye. Candidate alignment is the step that carries the policy across the dashed frozen-WM Top-1 reference $L_0 = 24.2\%$ (+7.8 pp), and Retrospection preserves that position.

among the world-model proposals.

Useful Evidence, Not More Evidence

We finally ask whether a generated trace contains decision-relevant state or only persuasive wording. Mentioning prior-action patterns becomes more frequent under GT-only supervision (22.4% $\rightarrow$ 37.8%), yet the accuracy gap associated with such mentions collapses from +13.5 pp to +3.2 pp. Candidate alignment also increases mention frequency to 32.4%, but preserves a much larger +12.3 pp evidence utility. The final EGO checkpoint mentions such evidence less often (17.5%) but raises its conditional utility to +18.3 pp. Thus, the central contrast is not whether a model *says* that it used temporal evidence, but whether the cited evidence separates correct from incorrect decisions.

The belief channel gives a complementary control. Under answer-only supervision, the task belief becomes identical to the selected action in 28.8% of traces, versus 0.0% for the base and 2.1% after candidate alignment; the final EGO model also has 0.0% echo. Yet swapping belief while holding the reasoning text fixed changes the selected action in 9.5% of candidate-aligned examples, versus 0.25% under the paraphrase self-control; subtracting that control gives a belief-swap sensitivity of 9.3%. Taken together, low echo and nonzero interventional sensitivity show that the belief field can carry decision-relevant state without degenerating into an action restatement. We do not interpret the observational

evidence split as causal; the paired belief swap supplies the intervention only for the belief claim. Table 3 shows one anticipation point end to end, where the echoed belief and the corrected one are visible in the generated traces.

What Is and Is Not Established

The current same-cohort evidence establishes three claims. First, candidate-constrained training improves selection beyond both the world-model Top-1 policy and answer-only supervision. Second, the improvement appears in both retention and correction rather than in a single easy subset. Third, the gain is causally dependent on completed-action history. Together these results support the paper’s modular account: the world model defines a visually grounded action boundary, while the language policy learns to interpret task and trajectory context within that boundary.

The final training stage augments Retrospection-with-Replay with grounding oversampling. Grounded examples increase from 12.8% to 32.0% at a fixed training-set size, preserving the optimization budget. The resulting EGO model reaches SelAcc 28.8%, G_1 42.5%, G_2 25.6%, continuation recall 48.4%, evidence utility +18.3 pp, and belief echo 0.0%, recovering most of the selection accuracy lost at the preceding Retrospection-with-Replay stage (26.4%). It nonetheless fails the preregistered non-inferiority margin of 1 pp against θ_{CE} : on the official paired set

Capability metric	Base	GT-only	Cand.-CE	EGO	What it tests
SelAcc@Top- K (% , $\uparrow$)	21.0	21.9	28.8	28.8	overall grounded selection
G_1 : retain WM-correct (% , $\uparrow$)	29.7	26.5	37.4	42.5	preserve a valid predictive prior
G_2 : correct WM-wrong (% , $\uparrow$)	18.7	20.8	27.8	25.6	reselect beyond WM Top-1
Continuation recall (% , $\uparrow$)	28.4	32.9	43.5	48.4	recover trajectory-consistent action
Continuation precision (%)	72.7	71.3	71.1	68.2	signal reliability (control)
Evidence utility (Δ accuracy, pp, $\uparrow$)	+13.5	+3.2	+12.3	+18.3	evidence-to-decision coupling
Evidence mention rate (%)	22.4	37.8	32.4	17.5	surface frequency (control)
Belief-action echo (% , $\downarrow$)	0.0	28.8	2.1	0.0	belief content beyond action copy

Table 1: **Capability profile on the common covered set ($n=1,000$)**. Columns correspond to the unaligned policy, the answer-only control, the candidate-aligned policy, and the final EGO model, respectively. Continuations form 31.0% of the set (310/1,000); their stable precision shows that the main limitation is recovery, not noisier continuation guesses. Evidence utility is the SelAcc difference between traces that do and do not explicitly invoke a prior-action pattern; it is observational, not a causal effect. Because the world-model pool covers 43.4% of held-out anticipation points, the full-set equivalent of 28.8% is 12.5%, against 10.5% for the WM Top-1 policy. The final EGO checkpoint is measured in the same cohort; its SelAcc uses the strict denominator, while the conditional capability axes exclude malformed traces following the definitions above.

Mechanistic diagnostic	Base	GT-only	Cand.-CE	EGO
Evidence mention (%)	22.4	37.8	32.4	17.5
Acc. when mentioned (%)	31.8	24.1	38.4	44.7
Acc. when absent (%)	18.4	21.0	26.1	26.4
Evidence utility (pp)	+13.5	+3.2	+12.3	+18.3
Belief-action echo (%)	0.0	28.8	2.1	0.0
Belief-swap sensitivity	0.073	0.085	0.093	—

Table 2: **Surface and interventional diagnostics**. Evidence utility is a conditional association and is interpreted descriptively. Belief-swap sensitivity is interventional: only the belief prefix is swapped while reasoning is held fixed, and the paraphrase-control flip rate is subtracted ($n=400$ for Base, GT-only, and Cand.-CE). No belief-swap evaluation was run for the final EGO model, so its entry is left blank rather than transferred from its predecessor. The low echo rate under candidate alignment shows that its belief field is not merely a copy of the selected action.

($n=937$, 86 video clusters) SelAcc changes by -0.64 pp (95% CI $[-3.88, +2.91]$) and G_2 by -2.53 pp (95% CI $[-6.98, +2.29]$); both intervals contain zero, and a conventional 4 pp margin would be met, but that reading is post hoc. We therefore report the final model without claiming that Retrospection preserves Prospection without degradation.

Limitation of the final-stage intervention. Axis 5 is not established for the final EGO model because no belief-swap evaluation was conducted after grounding oversampling. The preceding Retrospection-with-Replay model reached belief-swap sensitivity 0.098, below the preregistered 0.20 threshold, while its reasoning-swap sensitivity was 0.483, the highest among the evaluated conditions. Thus, the interventional mass moved to the reasoning channel rather than disappearing.

The final belief-action echo is 0.0%, but echo alone is not an interventional test; the positive belief-utility diagnostics are therefore retained only as exploratory observations.

Excluded proxy. First-person phrasing is deliberately excluded from the capability profile. For the GT-only control it changes from 30.7% with candidates presented to 91.0% in generation without candidates, and within GT-only traces it does not positively predict correctness (16.1% for first-person openings vs. 29.7% for annotation-style openings). It is therefore a prompt- and template-sensitive surface form, not evidence of embodied reasoning. We likewise omit contrastive connectives: they occur in 0.3% of base traces and 1.9% of final-model traces, and their within-arm accuracy association is unstable between the replay and grounding stages (+19.2 pp versus -7.6 pp, respectively, with only 18 final-stage traces), which is consistent with small-sample noise. Neither proxy is used to support the paper’s claims.

Cooking anticipation case. Observation interval 3620.4–3628.4s; target time 3631.0s; 122 completed actions in the available history.
WM candidate set (shuffled; scores and ranks hidden from the policy): check heat (GT, WM rank 3) · flip flatbread · transfer flatbread · cook flatbread · press dough · roll dough · knead dough (WM rank 1) · sprinkle flour · shape dough · cut dough.
Completed-action history (last 8): ... → roll dough → flip ingredient → check heat → roll dough → check heat → flip ingredient → press dough → cook flatbread.

Model	reasoning → belief → action
Base	<i>“The current frame SHOWS the flatbread being cooked on the hot surface, and the person’s hand is positioned to flip it... Flipping the flatbread is the logical next step to ensure even cooking.”</i> BELIEF — Flipping the flatbread to cook the other side. → flip flatbread × [scene description via “shows”]
GT-only	<i>“The current frame SHOWS the flatbread being cooked on the hot surface... The most logical next step, given the context of cooking flatbread, is to flip it... This aligns with <u>the previous actions</u> of checking heat and flipping the flatbread.”</i> BELIEF — flip flatbread [action-copying belief] → flip flatbread × [scene description via “shows”]
Cand.-CE	<i>“The current frame SHOWS the flatbread being cooked on the hot surface, and the previous action was flipping the ingredient. The next logical step is to continue cooking the flatbread, as it is still on the fire and needs to be cooked through.”</i> BELIEF — Cook the flatbread on the fire. → cook flatbread × [scene description via “shows”]
EGO	<i>“I have just finished pressing the dough and am now observing the fire to assess the heat for cooking. The flames are visible and active, indicating the stove is ready. I need to ensure the heat is suitable before proceeding with the next step in cooking.”</i> BELIEF — I am in the middle of assessing the heat level for cooking the flatbread. → check heat ✓ [state-bearing belief; no explicit prior-action reference]

Table 3: **A state-bearing belief enables correction when an action-copying belief does not.** The four evaluated policies receive the same observation, completed-action history, shuffled candidate set, and prompt. Generated traces are reproduced verbatim, with ... marking omitted text. To avoid anecdotal selection, we apply a fixed criterion to the common covered set: all four traces must be valid, GT-only must copy its chosen action into the belief field and answer incorrectly, and EGO must answer correctly. This criterion identifies 31 of 915 points; the example shown is one of the 18 for which Cand.-CE is also incorrect. Underlined spans explicitly reference prior actions under the same rule used for the mention-rate analysis, whereas **bold** spans express the model’s current task state.